

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	B. Janani	Reg. No.:	192321100
Section / Batch:	Slot A - 2023	Date:	12/8/26

Code of Conduct

I B. Janani (192321100) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Janani

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Terrance Frederick Fernandez	Terrance Frederick Fernandez	
Signature: <u>[Signature]</u>	Signature: <u>[Signature]</u>	Signature: _____
Date: <u>12/8/26</u>	Date: <u>12/8/26</u>	Date: _____

95
100

ITA 0301 Mobile Computing

B. JANANI
192321100

1) Given:

C03 AT3 Hackathon

Data Packet size = 500 bytes

Tunnel Overhead = 20 bytes per packet

Additional overhead = 20 bytes

Total transmitted packet size = $500 + 20 = 520$ bytes

Additional overhead = 20 bytes

2) Bandwidth = 20 MHz

Modulation = 64 QAM

Bits per symbol = 6 bits

Data Rate = Bandwidth $\times$ Bits per symbol

$$= 20 \times 10^6 \times 6$$

$$= 120 \times 10^6 \text{ bits/sec}$$

120 Mbps

QAM $\rightarrow$ Quadrature Amplitude Modulation - used to transmit digital data by changing both the amplitude & phase of a carrier signal.

Uses: $\rightarrow$ Wi-Fi network

$\rightarrow$ Used in 4G & 5G communication

$\rightarrow$ Used in Cable TV

3) No. of routing table entries = 50

Memory per entry = 32 bytes

Total memory = No. of entries $\times$ Memory per entry

$$= 50 \times 32 = 1600 \text{ bytes}$$

Proactive routing - The network keeps routes to all nodes ready all the time, even if you are not currently sending data. It updates the routing table

Eg: DSDV, OLSR

Reactive routing - The network finds a route only when it is needed.

Eg: AODV, DSR

4) Speed of mobile node = 10 m/s

Routing update interval = 2 secs

Distance = Speed $\times$ time

$$= 10 \times 2 = \boxed{20 \text{ m}}$$

5) Given;

Intermediate nodes = 5

Source node sends the first route request

$$\text{Total RREQ packets} = 1 + 5 = \boxed{6 \text{ route request packets}}$$

6) Subnet mask = $255.255.255.240$

This is a $/28$ subnet

$$\text{No. of host bits} = 32 - 28 = 4$$

$$\text{Total IP address} = 2^4 = 16$$

But 2 addresses cannot be assigned

→ 1 for Network add

→ 1 for Broadcast address

$$= 16 - 2 = \boxed{14}$$

7) Channel Bandwidth = 20 MHz

$$\text{Symbol rate} = 1 \text{ Mbps} = 1 \times 10^6 \text{ symbol/s}$$

$$\text{Spectral Efficiency} = \frac{\text{Symbol rate}}{\text{Bandwidth}}$$

$$= \frac{1 \times 10^6}{20 \times 10^6} = 0.05$$

Mbps - Mega Symbol per second

⇓

A symbol can carry one or more bits depending on the modulation.

$$\text{No. of Nodes} = 50$$

$$\text{Area} = 1 \text{ km}^2$$

$$\text{Network Density} = \frac{\text{No. of Nodes}}{\text{Area}}$$

$$= \frac{50}{1} \\ = 50 \text{ nodes / km}^2$$

$$\text{Network Density} = \frac{\text{No. of nodes} \times \pi r^2}{\text{Area}}$$

$$r = 250 \text{ m}, \text{ Area} = 1 \text{ km}^2 = 10^6 \text{ m}^2$$

$$= \frac{50 \times \pi \times (250)^2}{10^6} = \frac{50 \times 3.14 \times 62500}{10^6} = \frac{9817500}{10^6} = \boxed{9.82}$$

9) Speed of mobile node $v = 30 \text{ m/s}$

$$\text{Carrier frequency } f = 900 \text{ MHz} = 900 \times 10^6 \text{ Hz}$$

$$\text{Speed of light} = 3 \times 10^8 \text{ m/s}$$

$$f_d = \frac{v}{c} \times f \\ = \frac{30}{3 \times 10^8} \times 900 \times 10^6 = \boxed{90 \text{ Hz}}$$

$\Rightarrow$ Doppler shift - change in the received frequency of a signal because the transmitter or receiver is moving.

10) Data rate = 54 Mbps

$$\text{Packet size} = 1500 \text{ bytes}$$

$$\text{Packet size} \rightarrow \text{bits}$$

$$1500 \times 8 = 12000 \text{ bits}$$

$$\text{Transmission Time} = \frac{\text{Packet Size}}{\text{Data Rate}}$$

$$= \frac{12000}{54 \times 10^6}$$

$$= 0.0002222 \text{ s}$$

$$= 0.222 \text{ ms}$$

$$= 222 \mu\text{s}$$